

Finite-temperature GWR / Relaxed-core PAW

James BOUST^{1,2}

¹CEA DAM-DIF, F-91297 Arpajon, France

²Université Paris-Saclay, CEA, LMCE, 91680 Bruyères-le-Châtel, France

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Outline

- Finite-temperature GWR

- Relaxed-core PAW



GW in Real-space (GWR)

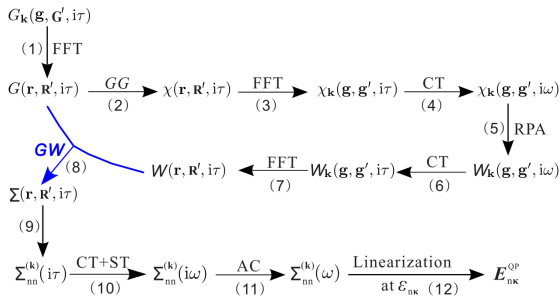
GW vs DFT

- More accurate excited states (e.g. gaps)
- More costly

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1. P. Liu *et al.*, Phys. Rev. B. (2016)

Figure: GWR algorithm workflow¹

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In ABINIT (M. Giantomassi)

- Zero temperature only
- Optimized imaginary grids (minimax)²

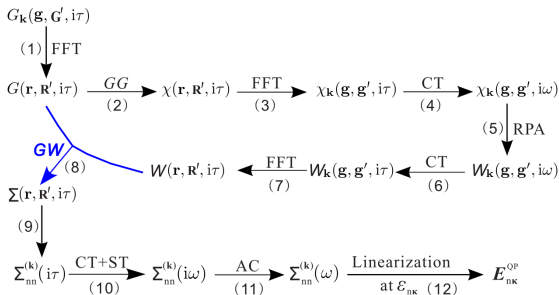


Figure: GWR algorithm workflow¹

1. P. Liu *et al.*, Phys. Rev. B. (2016)

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Non-zero temperature

- Fermi-Dirac statistics
- Finite-T optimized imaginary grids

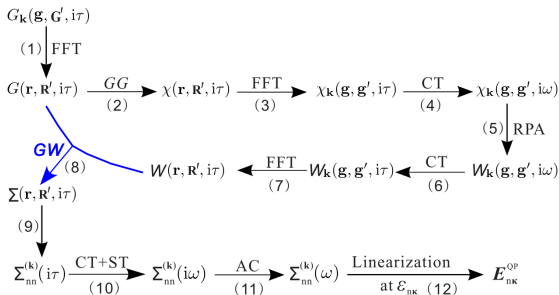


Figure: GWR algorithm workflow¹

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Finite-T grids for GWR

For bosonic time quantities at $\beta = 1$, $G(\tau) = \int_0^{\epsilon_{\max}} \langle \epsilon | \tau \rangle \rho(\epsilon) d\epsilon$ with $\langle \epsilon | \tau \rangle = \frac{\cosh\left(\frac{\epsilon}{2}(1-2|\tau|)\right)}{2 \cosh\left(\frac{\epsilon}{2}\right)}$.

Minimax^{1,2}

■ Minimax problem

$$\min_{\sigma_k, \tau_k} \max_{\epsilon} \left| \mathbb{1} - \sum_{k=1}^N \sigma_k |\tau_k\rangle \langle \tau_k| \right| \quad (1)$$

$$\langle \epsilon | \epsilon \rangle = \frac{\tanh\left(\frac{\epsilon}{2}\right)}{4\epsilon} + \frac{1 - \tanh^2\left(\frac{\epsilon}{2}\right)}{8} \quad (2)$$

- Matsubara sums (RPA,...)
- Limited accuracy

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Discrete Lehmann representation^{3,4} (DLR)

- Chebyshev interp. + interp. decomposition

$$\langle \epsilon | \tau \rangle \approx \sum_{j=1}^N \sum_{k=1}^N l_j(\epsilon) \langle \epsilon_j | \tau_k \rangle l_k(\tau) \quad (3)$$

$$G(\tau) \approx \sum_{j=1}^N \langle \epsilon_j | \tau \rangle g_j \quad (4)$$

- Evaluation on any grid point
- More expensive approach

Benchmark at low temperatures on Si

- GW improves gap
- Finite-T approaches agree with zero-T
- GWR needs only a few grid points (~ 20 vs ~ 1000)

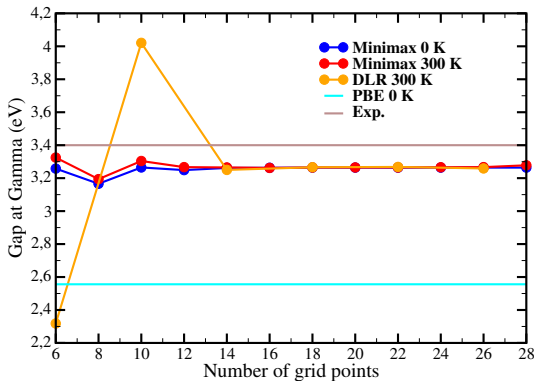


Figure: Calculated and experimental¹ Si gap at Γ .

Benchmark at high temperatures on Si

- Coherent with literature¹
- Further benchmarks required
- DFT $\neq$ GW

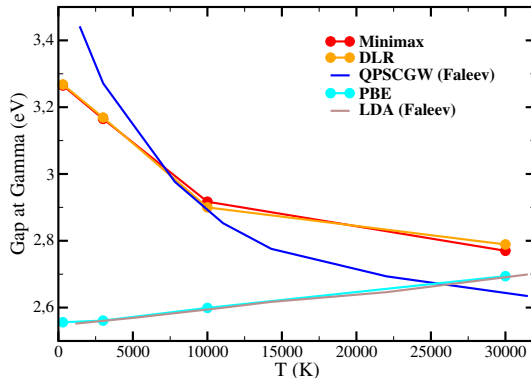


Figure: Calculated Si gap at Γ .

Conclusion

Summary

- Implemented finite-T GWR
- Tested two finite-T optimized imaginary grids for GWR (minimax and DLR)

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Perspectives

- Optimize and clean the code
- Try Nevanlinna¹ and Minimal Pole Model² analytic continuations
- Implement GWR+PAW

Outline

- Finite-temperature GWR

- Relaxed-core PAW



Frozen-Core Projector Augmented-Wave (FC-PAW)



Core states (ATOMPAW¹)

$$\begin{aligned} H_{\text{atom}}[n_{\text{atom}}] |\phi_i\rangle &= \epsilon_i |\phi_i\rangle \\ \updownarrow \\ n_{\text{atom}} &= \sum_i^{\text{occ.}} |\phi_i|^2 \end{aligned}$$

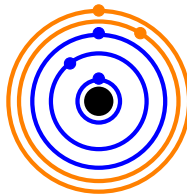
$n_c, \{\phi_i\}, \{\tilde{\phi}_i\}, \dots$

$$\begin{aligned} H[n_v + n_c] |\Psi_{n\mathbf{k}}\rangle &= \epsilon_{n\mathbf{k}} |\Psi_{n\mathbf{k}}\rangle \\ \updownarrow \\ n_v &= \sum_{n\mathbf{k}} f_{n\mathbf{k}} |\Psi_{n\mathbf{k}}|^2 \end{aligned}$$

Valence states (ABINIT²)

Frozen-Core

- Core frozen as in the isolated atom at $T = 0$ K
- Radial KS equation
- Low computational cost



Nucleus, core, valence (Al).

1. N. A. W. Holzwarth *et al.*, *Comput. Phys. Commun.* (2001)
2. M. J. Verstraete *et al.*, *J. Chem. Phys.* (2025)

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PAW transformation³ for valence states

- $|\Psi_{n\mathbf{k}}\rangle = |\tilde{\Psi}_{n\mathbf{k}}\rangle + \sum_i \langle \tilde{p}_i | \tilde{\Psi}_{n\mathbf{k}} \rangle (|\phi_i\rangle - |\tilde{\phi}_i\rangle)$
- $\tilde{H} = -\frac{1}{2}\Delta + \tilde{v}_{\text{eff}} + \sum_{ij} |\tilde{p}_i\rangle D_{ij} \langle \tilde{p}_j|$

1. N. A. W. Holzwarth *et al.*, *Comput. Phys. Commun.* (2001)
2. M. J. Verstraete *et al.*, *J. Chem. Phys.* (2025)
3. P. E. Blöchl, *Phys. Rev. B* (1994)

Limitations of FC-PAW in extreme conditions

No impact of temperature on core

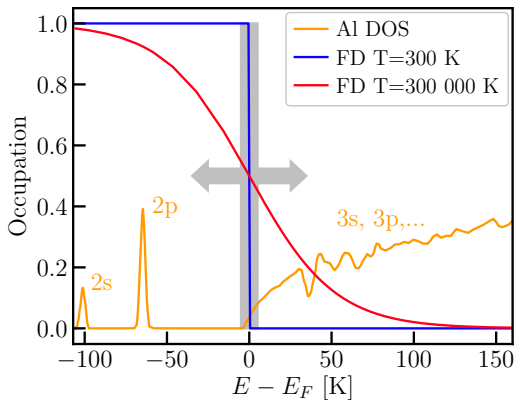
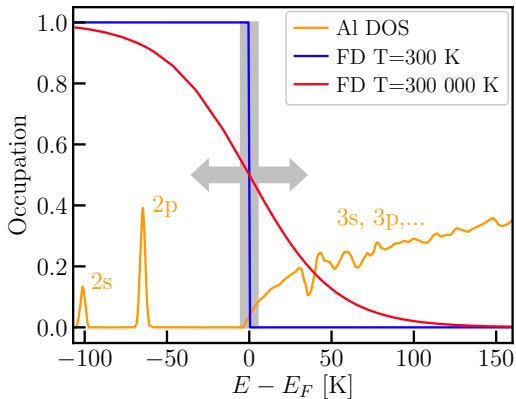


Figure: Al DOS and Fermi-Dirac occupation.

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High energy orbitals: ext.-FPMD^{1,2}

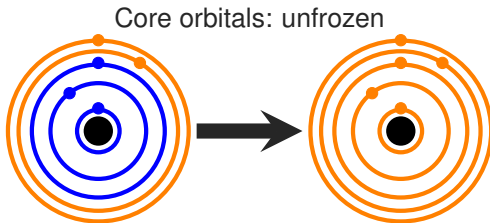


Figure: Al DOS and Fermi-Dirac occupation.

1. S. Zhang *et al.*, Phys. of Plasmas (2013)
2. A. Blanchet *et al.*, Comput. Phys. Commun. (2022)

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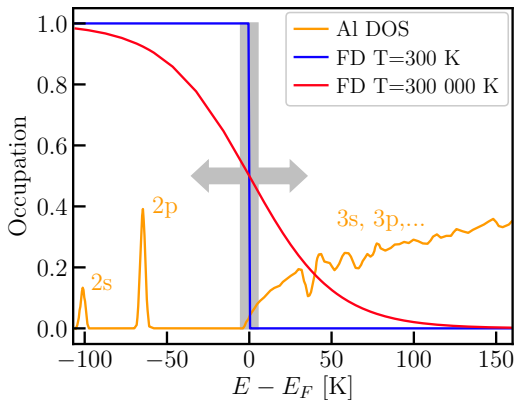


Figure: Al DOS and Fermi-Dirac occupation.

No impact of environment on core

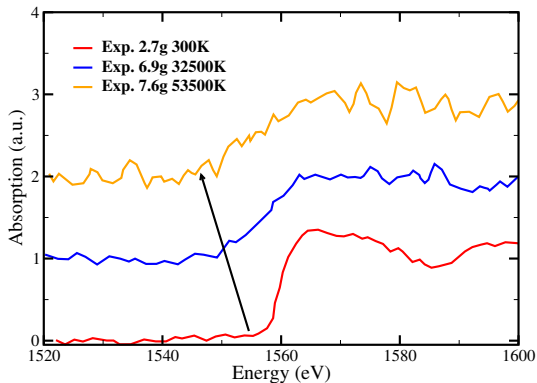
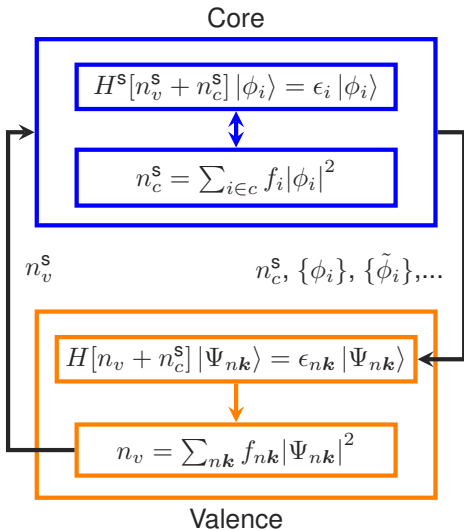


Figure: Experimental³ K-edge Al XANES spectrum for reshocked conditions.

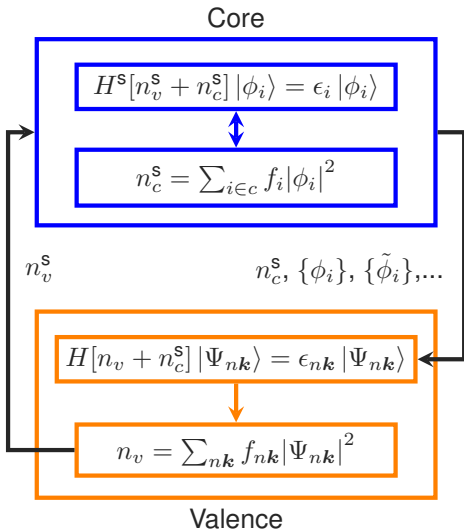
Relaxed-Core PAW (RC-PAW) in ABINIT



Finite-T generalization of RC-PAW from Marsman¹

- Self-consistent core but spherical symmetry
- Low added computational cost

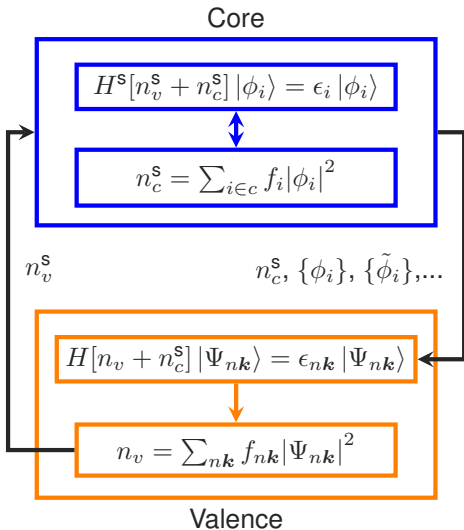
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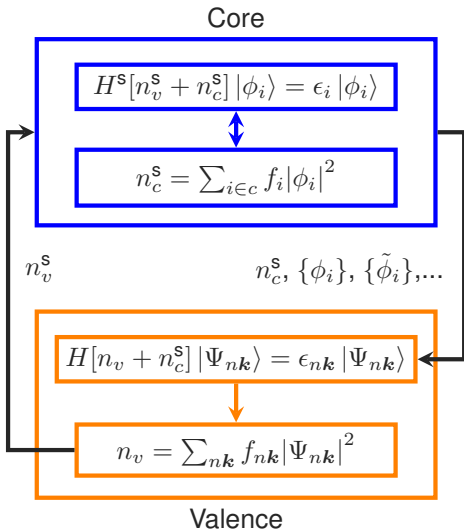
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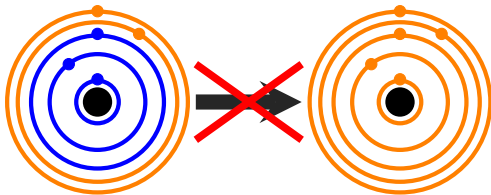
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- Correct core eigenenergies + Fermi-Dirac statistics

Relaxed-Core PAW (RC-PAW) in ABINIT



Finite-T generalization of RC-PAW from Marsman¹

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- Core in sphere still required
- Correct core eigenenergies + Fermi-Dirac statistics



Application to aluminium at high pressures

- Correctly captures the shift due to compression
- Probably requires core-hole effect

Calculation details

- RC-PAW
- ext.-FPMD
- MLACS¹ NVT
- CONDUCTI for σ_1

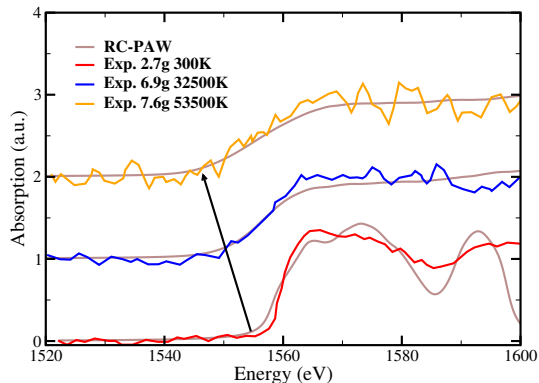


Figure: Calculated and experimental² K-edge Al XANES spectrum for reshocked conditions.

Application to molybdenum at high temperatures (with A. Blanchet)

- RC-PAW in excellent agreement
- *ecut* much lower for RC-PAW
- AE too expensive in MD for large boxes

Calculation details

- BCC Mo
- $T = 3 \cdot 10^7$ K ($1s$ ionized)
- ext.-FPMD

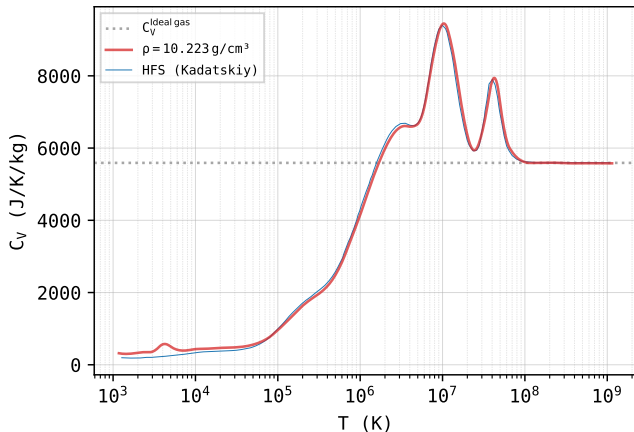
	RC-PAW ($1s^2$)	AE
Free energy (Ha)	66106.2	66105.9
Pressure (GPa)	$1.1767 \cdot 10^6$	$1.1765 \cdot 10^6$
Fermi level (Ha)	-658.8	-658.8
<i>ecut</i> (Ha)	100	500
Wall time (min.)	6	29

Application to molybdenum at high temperatures (with A. Blanchet)

Excellent agreement
with average atom.

Calculation details

- RC-PAW
- $\rho = \rho_0$
- Isokinetic MD
- ext.-FPMD



Conclusion

Summary

- Implemented and benchmarked finite-T RC-PAW
- Managed to capture temperature and environment effects on core without the cost of AE
- Applied to Al and Mo in extreme conditions

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- Managed to capture temperature and environment effects on core without the cost of AE
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Perspectives

- Manuscript in preparation
- Dirac equation

Benchmark at low temperatures on Si

Finite-T approaches agree with zero-T.

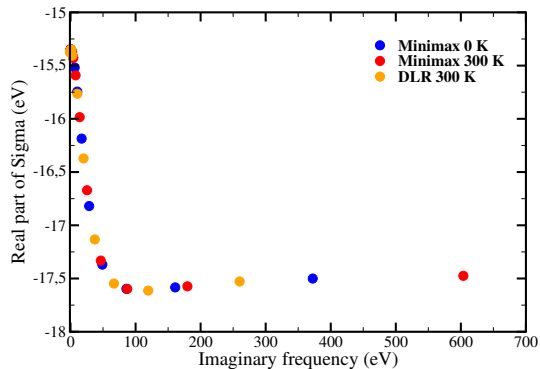


Figure: Real part of the self-energy for one band of Si at Γ .

Application to aluminium at high pressures

Ne core ($1s^2 2s^2 2p^2$) vs all-electron (AE)

- FC-PAW fails
- RC-PAW agrees with AE

Calculation details

- FCC Al
- $T = 300$ K
- CONDUCTI for σ_1

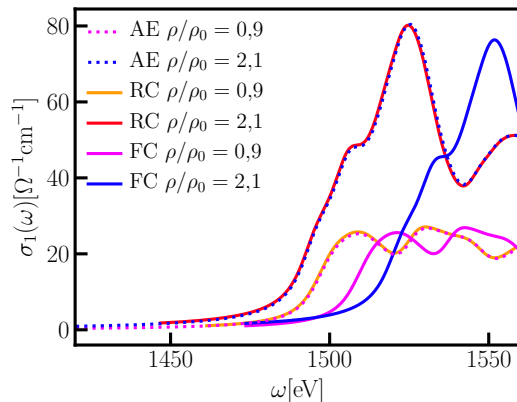


Figure: Calculated K-edge Al XANES spectrum for different densities ρ .